

polya-neumann-balls.nb

Scripts for the paper “Pólya’s conjecture for higher-dimensional Neumann balls”

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The sections appear in a slightly different order to those in the paper

Out[272]=

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Programming extras

In[160]:=

```
SetDirectory[NotebookDirectory[]];
clr = ColorData[97, "ColorList"];
<< MaTeX`
SetOptions[MaTeX,
  "BasePreamble" → {"\\usepackage[lining]{ebgaramond}", "\\usepackage{amsmath}",
    "\\usepackage[ebgaramond]{newtxmath}", "\\usepackage{bm}"},
  FontSize → 12, Magnification → 1];
```

In[28]:= ? CenteredInterval

```

LUBounds[expr_, eps_] := Information[CenteredInterval[expr, eps], "Bounds"];
LBound[expr_, eps_] := LUBounds[expr, eps][[1]];
UBound[expr_, eps_] := LUBounds[expr, eps][[2]];

```

Out[28]=

Symbol

CenteredInterval $[x, dx]$ for real numbers x and dx gives a centered interval that contains the real interval $\{a \in \mathbb{R} \mid x - dx \leq a \leq x + dx\}$.

CenteredInterval $[x + i y, dx + i dy]$ gives a centered interval that contains the complex rectangle $\{a + i b \in \mathbb{C} \mid x - dx \leq a \leq x + dx \wedge y - dy \leq b \leq y + dy\}$.

CenteredInterval $[c]$ for an approximate number c gives a centered interval that contains all values within the error bounds of c .

§3. Large λ

Various definitions and illustrations

■ Definitions from Proposition 3.3

```

In[5]:= t1_0[d_] := (1 + 8 d + 32 d^2) / (64 d^(5/2)); t1_3[d_] := If[d <= 6, 0, -6 + d] / (6 + 24 d);

```

```

T1[d_, tau_] := t1_0[d] + t1_3[d] tau^(-3);

```

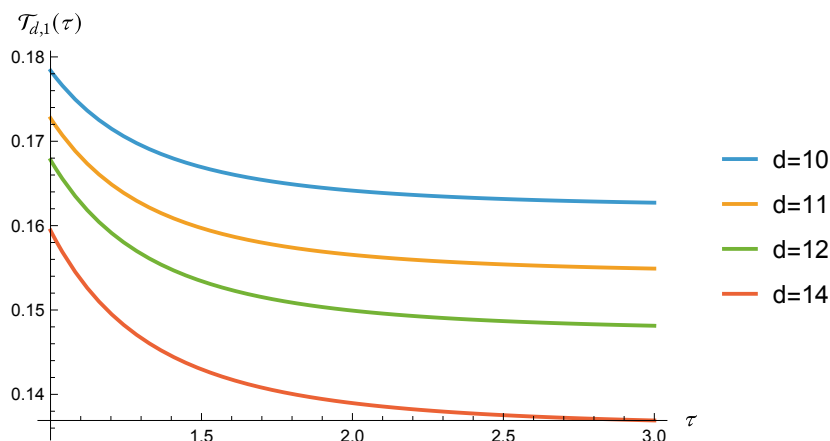
■ Extra illustration, typical plots of $T_1[d, \tau]$:

```

In[22]:= Plot[Table[T1[d, tau], {d, {10, 11, 12, 14}}] // Evaluate, {tau, 1, 3},
  AxesLabel -> {MaTeX["\\tau"], MaTeX["\\mathcal{T}_1[d,1](\\tau)"]},
  PlotLegends -> Table["d=" <> ToString[d], {d, {10, 11, 12, 14}}]]

```

Out[22]=



■ Definitions from Proposition 3.4

$$\text{In[16]:= } t_{2,2} = \frac{3^{2/3} \pi^{7/6}}{20 \times 2^{5/6}};$$

$$\alpha_4[d_] := \frac{(3\pi)^{2/3} (14\sqrt{2} - 10\sqrt{2}d + 3\pi)}{56 \times 2^{5/6} d};$$

$$\alpha_6[d_] := -\frac{(-2+d)\pi^{4/3} (18 - 10d + 3\sqrt{2}\pi)}{64 \times 6^{2/3} d^2};$$

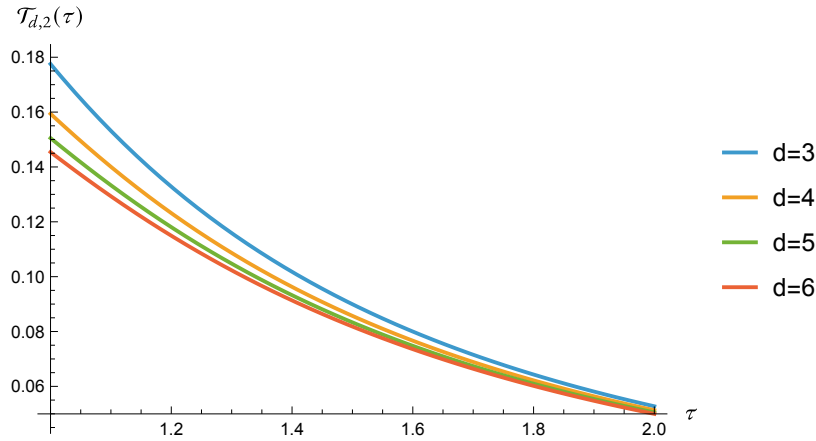
$$\alpha_8[d_] := \frac{27(-3+d)(-2+d)\pi^2 (-4 + \sqrt{2}\pi)}{11264 d^3};$$

$$\text{In[20]:= } T_2[d_, \tau_] := t_{2,2} (\tau^{(-2)} + \alpha_4[d] \tau^{(-4)} + \alpha_6[d] \tau^{(-6)} + \alpha_8[d] \tau^{(-8)});$$

■ Extra illustration, typical plots of $T_2[d, \tau]$:

$\text{In[21]:= Plot[Table}[T_2[d, \tau], \{d, \{3, 4, 5, 6\}\}] // \text{Evaluate}, \{\tau, 1, 2\},$
 $\text{AxesLabel} \rightarrow \text{MaTeX}[\{"\tau", "\mathcal{T}_2[d, 2](\tau)"}],$
 $\text{PlotLegends} \rightarrow \text{Table}["d=" <> \text{ToString}[d], \{d, \{3, 4, 5, 6\}\}]]$

Out[21]=



■ Definitions from Proposition 3.5

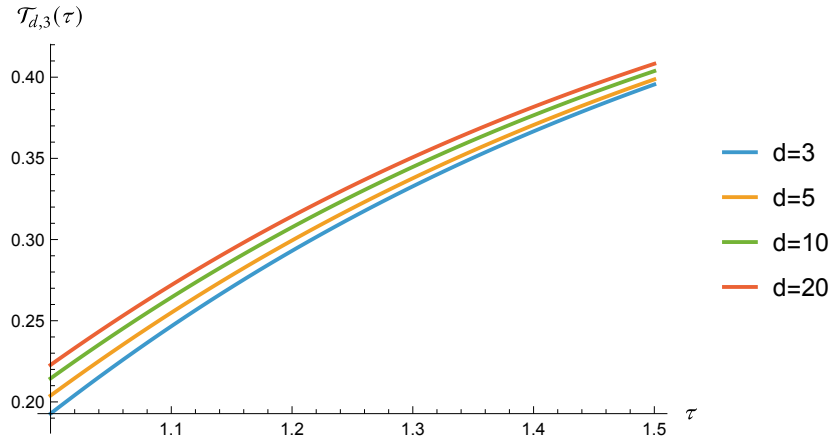
$$\text{In[23]:= } \omega = \frac{(3\pi)^{2/3}}{4 \times 2^{1/3}};$$

$$T_3[d_, \tau_] := \text{Sqrt}[\text{Pi} / 8] e^{-\frac{\omega}{\tau^2}} \left(1 - \frac{(-6+d)(-2+d)(-1+d)}{24 d (d \tau^3 - \tau \omega)^2} - \frac{d-1}{2 \sqrt{d} (d \tau^3 - \tau \omega)} \right);$$

■ Extra illustration, typical plots of $T_3[d, \tau]$:

```
In[25]:= Plot[Table[T3[d, τ], {d, {3, 5, 10, 20}}] // Evaluate, {τ, 1, 1.5},
  AxesLabel → MaTeX[{"\\tau", "\\mathcal{T}_{d,3}(\\tau)"}],
  PlotLegends → Table["d=" <> ToString[d], {d, {3, 5, 10, 20}}]]
```

Out[25]=



■ Definition of $\mathcal{T}_d$

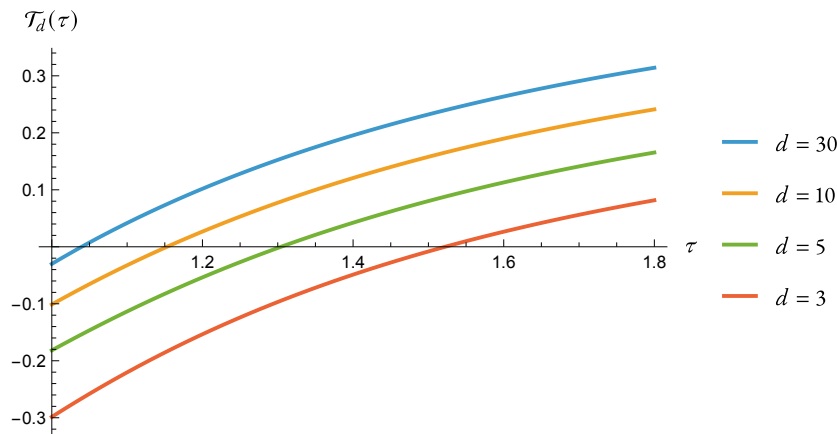
```
In[26]:= T[d_, τ_] := T3[d, τ] - T1[d, τ] - T2[d, τ];
```

■ Figure 3

In[134]=

```
figTdtypical = Plot[{T[30, τ], T[10, τ], T[5, τ], T[3, τ]},
  {τ, 1, 1.8}, AxesLabel → MaTeX[{"\\tau", "\\mathcal{T}_d(\\tau)"}],
  PlotLegends → MaTeX[Reverse[{"d=3", "d=5", "d=10", "d=30"}]]]
```

Out[134]=



In[135]=

```
Export["figTdtypical.pdf", figTdtypical]
```

Out[135]=

figTdtypical.pdf

Rational approximations for bounds in Theorem 3.1

■ Proving that $\mathcal{T}_d(1)$ is negative for $d=60$ and positive for $d=61$

In[33]:= **bounds60 = LUBounds[T[60, 1], 10^(-4)]**

Out[33]=
$$\left\{ -\frac{44\,760\,445\,596\,801\,377\,675}{75\,557\,863\,725\,914\,323\,419\,136}, -\frac{29\,648\,872\,838\,231\,958\,923}{75\,557\,863\,725\,914\,323\,419\,136} \right\}$$

In[34]:= **bounds61 = LUBounds[T[61, 1], 10^(-4)]**

Out[34]=
$$\left\{ \frac{26\,358\,269\,750\,698\,667}{4\,722\,366\,482\,869\,645\,213\,696}, \frac{970\,831\,567\,161\,287\,339}{4\,722\,366\,482\,869\,645\,213\,696} \right\}$$

■ Proving that $\mathcal{T}_d(28/25)$ is negative for d=12 and positive for d=13

In[35]:= **bounds12 = LUBounds[T[12, 28/25], 10^(-3)]**

bounds13 = LUBounds[T[13, 28/25], 10^(-3)]

Out[35]=
$$\left\{ -\frac{25\,931\,230\,113\,289\,738\,127}{4\,722\,366\,482\,869\,645\,213\,696}, -\frac{16\,486\,497\,145\,626\,302\,351}{4\,722\,366\,482\,869\,645\,213\,696} \right\}$$

Out[36]=
$$\left\{ \frac{7\,605\,667\,427\,504\,885\,135}{9\,444\,732\,965\,739\,290\,427\,392}, \frac{26\,495\,133\,362\,831\,756\,687}{9\,444\,732\,965\,739\,290\,427\,392} \right\}$$

■ Numerical values of τ_d^*

```

τstar = <|
  Table[d → (SolveValues[{T[d, τ] == 0, 1 < τ < 17/10}, τ][[1]] // N), {d, 3, 60}]] >;
TableForm[Table[{d, τstar[d], d + 29, τstar[d + 29]}, {d, 3, 31}],
  TableHeadings → {None, MaTeX[{"d", "\\tau_d^*", "d", "\\tau_d^*"}]}]

```

Out[157]//TableForm=

d	τ_d^*	d	τ_d^*
3	1.52217	32	1.03525
4	1.38519	33	1.03323
5	1.30432	34	1.03131
6	1.24964	35	1.02947
7	1.21574	36	1.02772
8	1.18994	37	1.02605
9	1.16954	38	1.02444
10	1.15291	39	1.02291
11	1.13904	40	1.02143
12	1.12727	41	1.02002
13	1.11711	42	1.01866
14	1.10824	43	1.01734
15	1.10041	44	1.01608
16	1.09344	45	1.01486
17	1.08718	46	1.01369
18	1.08152	47	1.01255
19	1.07637	48	1.01146
20	1.07166	49	1.01039
21	1.06733	50	1.00937
22	1.06333	51	1.00837
23	1.05963	52	1.0074
24	1.05618	53	1.00647
25	1.05297	54	1.00556
26	1.04996	55	1.00468
27	1.04714	56	1.00382
28	1.04449	57	1.00299
29	1.04198	58	1.00217
30	1.03962	59	1.00138
31	1.03737	60	1.00062

§4. Small λ

Proposition 4.2 and Remark 4.3

```

In[98]:= τhash[d_] := 21/3 (1 + d) $\frac{1}{3d}$  Gamma[1 +  $\frac{d}{2}$ ] $\frac{2}{(3d)}$  / Sqrt[d];

```

```

In[99]:= τhash[3]

```

Out[99]=

$$\frac{(2\pi)^{1/9}}{3^{5/18}}$$

In[100]:=

```

LBound[τhash[3] - 17/20, 10-2]

```

Out[100]=

$$\frac{1\,661\,057\,252\,282\,206\,389\,201}{37\,778\,931\,862\,957\,161\,709\,568}$$

§4.3. Variational approach: the algorithm

■ General definitions

```
In[43]:= Iis[ρ_, d_] := {
  Integrate[D[ρ, x]^2 x^(d-1), {x, 0, 1}, Assumptions → d ≥ 3],
  Integrate[ρ^2 x^(d-3), {x, 0, 1}, Assumptions → d ≥ 3],
  Integrate[ρ^2 x^(d-1), {x, 0, 1}, Assumptions → d ≥ 3]
};
FdOfTau[d_, Iif_, τ_] := Module[{X, k, λ, Iid},
  Iid = Iif[d] // Evaluate;
  X = Sqrt[(λ^2 Iid[[3]] - Iid[[1]]) / Iid[[2]] + (d/2 - 1)^2] - (d/2 - 1);
  Sqrt[2 Pi] (1 + 1/(4 d)) d^(3/2) (X + (d - 3)/2)
  Product[X + k, {k, 0, d - 3}] / λ^d /. λ → τ^3 d^(3/2)
];
```

§4.4. Constant test function

■ Integrals

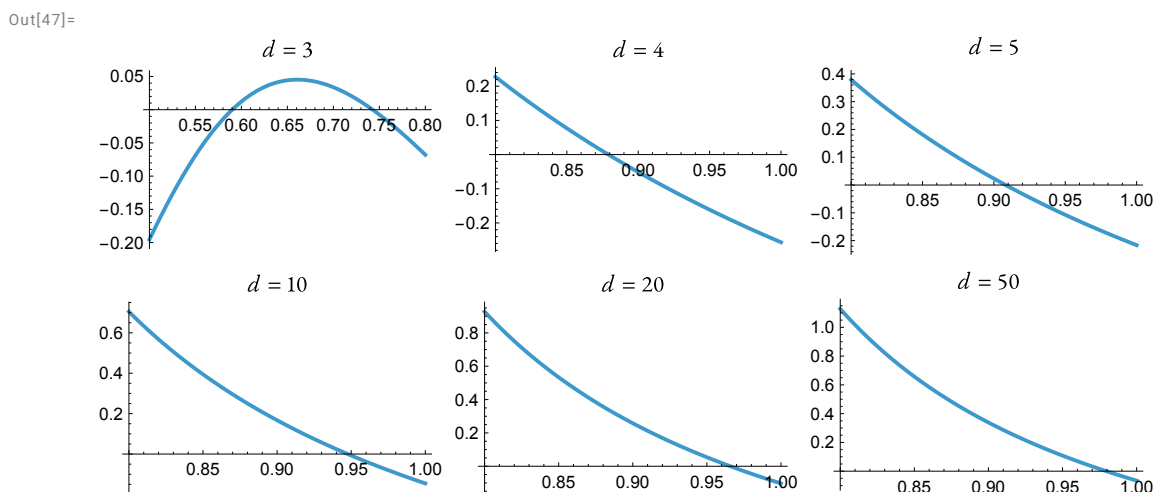
```
In[45]:= Ii1 = Iis[1, d]
Ii1f[d_] := Evaluate[Ii1];
```

Out[45]=

$$\left\{0, \frac{1}{-2+d}, \frac{1}{d}\right\}$$

■ Figure 4

```
In[47]:= figvar1 = Grid[ArrayReshape[
  Table[
    Plot[FdOfTau[d, Ii1f, τ] - 1, {τ, If[d == 3, 0.5, 0.8], If[d == 3, 0.8, 1.0]},
    ImageSize → Small, PlotLabel → MaTeX["d=" <> ToString[d]],
    {d, {3, 4, 5, 10, 20, 50}}, {2, 3}]]]
```



```
In[*]:= Export["figvarone.pdf", figvar1]
```

Out[*]=
figvarone.pdf

■ Lower bound for $\mathfrak{F}_7 \left(\tau^{\circ 3} 7^{3/2} \right)$

```
In[48]:= lowerFd7 = 
$$\frac{193 \left( 1 - \frac{1}{d(d-2)} \right) \left( 1 + \frac{(-2+d)(-1+d)}{8d^2 \tau^6} \right) \left( 1 - \frac{1}{24 \tau^6} - \frac{d-1}{2 \sqrt{-2+d} d \tau^3} \right)}{77 e \tau^3} /.$$

```

```
{d -> 7, \tau -> 17/20} // Simplify
```

```
lowerFd7 // N
```

```
LBound[lowerFd7 - 1, 10^(-2)]
```

```
Out[48]= 
$$-\frac{878\,684\,768\,105\,600 \left( -450\,888\,949 + 70\,747\,200 \sqrt{5} \right)}{93\,387\,562\,322\,467\,096\,652\,199 e}$$

```

```
Out[49]= 1.01312
```

```
Out[50]= 
$$\frac{14\,751\,612\,740\,489\,335\,089}{4\,722\,366\,482\,869\,645\,213\,696}$$

```

■ Lower bounds for $\mathfrak{F}_d \left(\tau^{\circ 3} d^{3/2} \right)$, d=4, 5, 6

```
In[51]:= TableForm[Table[Flatten[{d, FdOfTau[d, IIf, 17/20] // N,
    LBound[FdOfTau[d, IIf, 17/20] - 1, 10^(-2)]}], {d, 4, 6}],
    TableHeadings -> {None, MaTeX[{"d", "\text{numerical}
    }\mathfrak{F}_d\left(\tau^{\circ 3} d^{3/2}\right)",
    "\text{lower bounds for }\mathfrak{F}_d\left(\tau^{\circ 3} d^{3/2}\right)-1\text{, all positive}"}]]]
```

```
Out[51]//TableForm=
```

d	numerical $\mathfrak{F}_d \left(\tau^{\circ 3} d^{3/2} \right)$	lower bounds for $\mathfrak{F}_d \left(\tau^{\circ 3} d^{3/2} \right) - 1$, all positive
4	1.07694	$\frac{5\,058\,082\,967\,777\,020\,815\,877}{75\,557\,863\,725\,914\,323\,419\,136}$
5	1.1817	$\frac{25\,947\,061\,976\,658\,208\,927\,419}{151\,115\,727\,451\,828\,646\,838\,272}$
6	1.24933	$\frac{36\,165\,778\,212\,213\,842\,916\,809}{151\,115\,727\,451\,828\,646\,838\,272}$

■ τ_d°

```
In[85]:= \tcirc = Association[
    Table[d -> \tau /. FindRoot[FdOfTau[d, IIf, \tau] - 1, {\tau, 1}], {d, 4, 63}]];
```



```
In[87]:= TableForm[Table[{d,  $\tau_{\text{circ}}[d]$ , d + 20,  $\tau_{\text{circ}}[d + 20]$ , d + 40,  $\tau_{\text{circ}}[d + 40]$  },
  {d, 4, 23}], TableHeadings → {None,
  MaTeX[{"d", "\\tau_d\\circ", "d", "\\tau_d\\circ", "d", "\\tau_d\\circ"}]]
```

Out[87]//TableForm=

d	τ_d°	d	τ_d°	d	τ_d°
4	0.879101	24	0.969255	44	0.978072
5	0.908255	25	0.96996	45	0.97834
6	0.922976	26	0.97062	46	0.9786
7	0.932252	27	0.97124	47	0.97885
8	0.938794	28	0.971823	48	0.979093
9	0.943735	29	0.972373	49	0.979327
10	0.947643	30	0.972892	50	0.979554
11	0.950836	31	0.973385	51	0.979774
12	0.953511	32	0.973852	52	0.979988
13	0.955795	33	0.974296	53	0.980194
14	0.957776	34	0.974719	54	0.980395
15	0.959516	35	0.975122	55	0.980591
16	0.961059	36	0.975508	56	0.98078
17	0.962442	37	0.975876	57	0.980965
18	0.963689	38	0.976229	58	0.981144
19	0.964822	39	0.976567	59	0.981319
20	0.965858	40	0.976892	60	0.981489
21	0.966809	41	0.977204	61	0.981654
22	0.967686	42	0.977504	62	0.981816
23	0.968499	43	0.977793	63	0.981973

§4.5. Alternative test function

■ Integrals

```
In[52]:=  $\rho_{\text{Alt}}[d_, x_] := \text{Piecewise}[\{\{0, 0 \leq x \leq 1 - 2/d\}\}, x^{(1-d)/2} (x - 1 + 2/d)^{(3/2)}];$ 
 $\text{IiAlt} = \text{Iis}[\rho_{\text{Alt}}[d, x], d] // \text{Simplify}$ 
 $\text{IiAltf}[d_] := \text{Evaluate}[\text{IiAlt}];$ 
```

Out[53]=

$$\left\{ -\frac{8 - 88d + 98d^2 - 42d^3 + 6d^4 + 3(-2 + d)^2 d (3 - 4d + d^2) \text{Log}\left[\frac{-2+d}{d}\right]}{4d^3}, \frac{-8 + 18d - 6d^2 - 3(-2 + d)^2 d \text{Log}\left[\frac{-2+d}{d}\right]}{d^3}, \frac{4}{d^4} \right\}$$

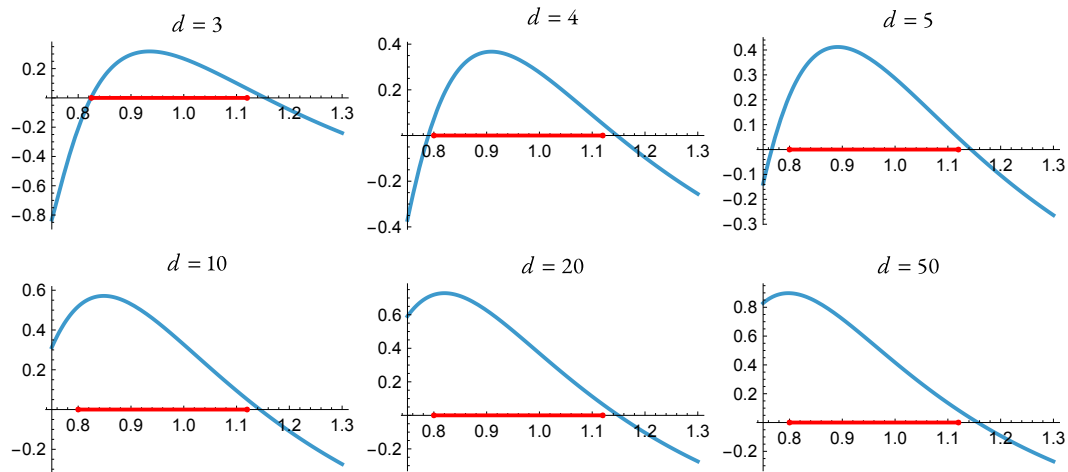
■ Figure 5

```

In[55]:= figvarNF = Grid[ArrayReshape[
  Table[
    Plot[FdOfTau[d, IiAltf,  $\tau$ ] - 1, { $\tau$ , 0.75, 1.3},
    ImageSize → Small, PlotLabel →  $\text{MaTeX}["d=" \<> \text{ToString}[d]]$ ,
    Epilog → {Red, Thick, Line[{If[d == 3, 33 / 40, 0.8], 0}, {1.12, 0}]},
    Point[{If[d == 3, {33 / 40, 0}, {0.8, 0}], {1.12, 0}]}]},
    {d, {3, 4, 5, 10, 20, 50}}], {2, 3}]]

```

Out[55]=



```

In[56]:= Export["figvarNF.pdf", figvarNF]

```

Out[56]=

figvarNF.pdf

■ Case d=3

```

In[75]:= f31 = FdOfTau[3, IiAltf, 33 / 40] // FullSimplify
f31 // N
LBound[f31 - 1, 10^(-3)]

```

Out[75]=

$$\frac{832\,000\sqrt{2\pi}\left(-16\,000 + \sqrt{256\,000\,000 + \frac{5\,991\,211\,721}{-8+9\log[3]}}\right)^2}{3\,759\,330\,236\,558\,193}$$

Out[76]=

1.00521

Out[77]=

$$\frac{39\,743\,541\,869\,798\,107\,149}{9\,444\,732\,965\,739\,290\,427\,392}$$

```
In[78]:= f32 = FdOfTau[3, IiAltf, 28 / 25] // FullSimplify
f32 // N
LBound[f32 - 1, 10 ^ (-3)]
```

```
Out[78]=
```

$$\frac{203\,125 \sqrt{\frac{\pi}{2}} \left(-15\,625 + \sqrt{244\,140\,625 + \frac{64\,021\,110\,026}{-8+9 \log[3]}} \right)^2}{6\,854\,839\,457\,808\,384}$$

```
Out[79]=
```

1.0633

```
Out[80]=
```

$$\frac{18\,829\,485\,080\,933\,663\,642\,335}{302\,231\,454\,903\,657\,293\,676\,544}$$

■ $\tau_{d,-}^{\bullet}$ and $\tau_{d,+}^{\bullet}$

```
In[94]:= \taubulletminus = Association[
  Table[d \to \tau /. FindRoot[FdOfTau[d, IiAltf, \tau] - 1, {\tau, If[d == 3, 0.8, 0.7]}][[1]],
    {d, 3, 62}]];
\taubulletplus = Association[
  Table[d \to \tau /. FindRoot[FdOfTau[d, IiAltf, \tau] - 1, {\tau, 1.2}][[1]], {d, 3, 62}]]];
```

```
In[97]:= TableForm[Table[{d,  $\tau_{d,-}$ ,  $\tau_{d,+}$ , d + 30,
 $\tau_{d+30,-}$ ,  $\tau_{d+30,+}$  }, {d, 3, 32}], TableHeadings →
{None,  $\tau_{d,-}$ ,  $\tau_{d,+}$ , d + 30,  $\tau_{d+30,-}$ ,  $\tau_{d+30,+}$ }]
```

Out[97]//TableForm=

d	$\tau_{d,-}$	$\tau_{d,+}$	d	$\tau_{d,-}$	$\tau_{d,+}$
3	0.82427	1.1545	33	0.654647	1.15355
4	0.789579	1.14687	34	0.653783	1.15378
5	0.766111	1.14398	35	0.652961	1.15401
6	0.748883	1.14303	36	0.652179	1.15422
7	0.735612	1.14293	37	0.651433	1.15443
8	0.725033	1.14322	38	0.650721	1.15464
9	0.71638	1.1437	39	0.650041	1.15483
10	0.709155	1.14426	40	0.64939	1.15502
11	0.703022	1.14485	41	0.648766	1.15521
12	0.697742	1.14544	42	0.648168	1.15539
13	0.693144	1.14602	43	0.647594	1.15556
14	0.689099	1.14659	44	0.647042	1.15573
15	0.68551	1.14713	45	0.646511	1.15589
16	0.682301	1.14764	46	0.646	1.15605
17	0.679412	1.14814	47	0.645508	1.15621
18	0.676797	1.1486	48	0.645034	1.15636
19	0.674416	1.14905	49	0.644576	1.1565
20	0.672239	1.14947	50	0.644134	1.15664
21	0.670239	1.14988	51	0.643707	1.15678
22	0.668395	1.15026	52	0.643294	1.15692
23	0.666688	1.15063	53	0.642894	1.15705
24	0.665103	1.15098	54	0.642507	1.15718
25	0.663627	1.15132	55	0.642133	1.1573
26	0.662248	1.15164	56	0.641769	1.15743
27	0.660957	1.15195	57	0.641417	1.15755
28	0.659745	1.15224	58	0.641075	1.15766
29	0.658606	1.15252	59	0.640743	1.15778
30	0.657531	1.1528	60	0.640421	1.15789
31	0.656517	1.15306	61	0.640107	1.158
32	0.655557	1.15331	62	0.639803	1.15811

Summary of low λ bounds

■ Figure 6

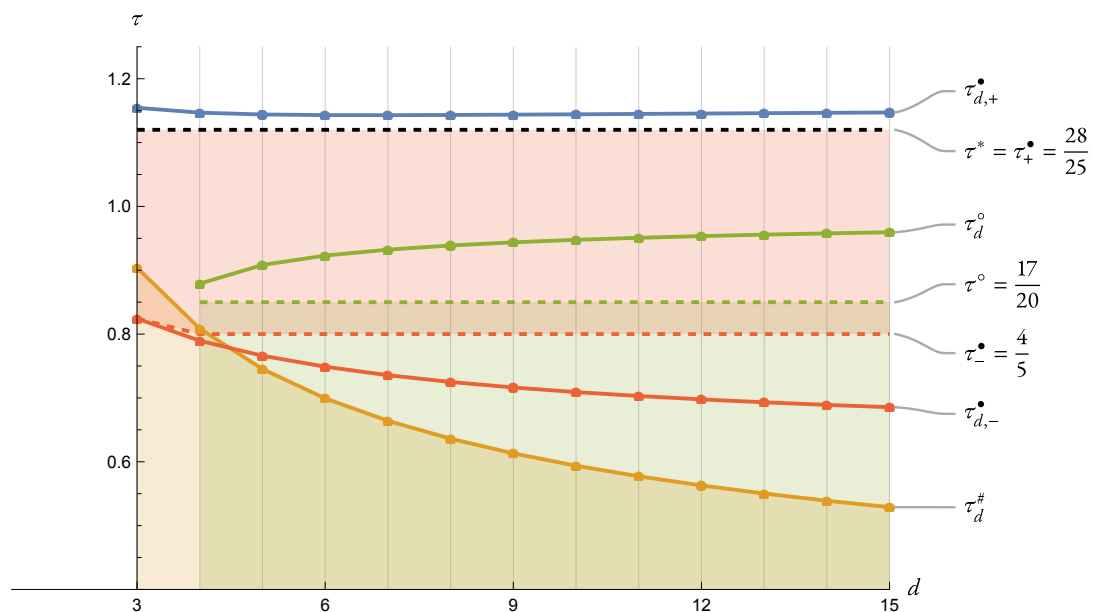
In[131]:=

```

dmax = 15;
figtotalallowpict = ListLinePlot[
  {Table[{d, 28/25}, {d, 3, dmax}], (*1*)
    Table[{d,  $\tau$ bulletplus[d]}, {d, 3, dmax}], (*2*)
    Table[{d,  $\tau$ hash[d]}, {d, 3, dmax}], (*3*)
    Table[{d,  $\tau$ circ[d]}, {d, 4, dmax}], (*4*)
    Table[{d, 17/20}, {d, 4, dmax}], (*5*)
    Table[{d,  $\tau$ bulletminus[d]}, {d, 3, dmax}], (*6*)
    Prepend[Table[{d, 4/5}, {d, 4, dmax}], {3,  $\tau$ bulletminus[3]}] (*7*)},
  PlotMarkers  $\rightarrow$  {
    None,
    { $\bullet$ , Offset[2]},
    { $\bullet$ , Offset[2]},
    { $\bullet$ , Offset[2]},
    None,
    { $\bullet$ , Offset[2]},
    None },
  PlotStyle  $\rightarrow$  {
    {Black, Thick, Dashed},
    {clr[[1]], Thick},
    clr[[2]],
    clr[[3]],
    {clr[[3]], Thick, Dashed},
    clr[[4]],
    {clr[[4]], Thick, Dashed}},
  PlotLabels  $\rightarrow$  {
    MaTeX["\\tau^*=\\tau^\\bullet_+=\\frac{28}{25}"],
    MaTeX["\\tau^\\bullet_{d,+}"],
    MaTeX["\\tau^\\bullet_{\\#_d}"],
    MaTeX["\\tau^\\bullet_{\\text{circ}_d}"],
    MaTeX["\\tau^\\bullet_{\\text{circ}}=\\frac{17}{20}"],
    MaTeX["\\tau^\\bullet_{d,-}"],
    MaTeX["\\tau^\\bullet_{-}=\\frac{4}{5}"]},
  Filling  $\rightarrow$  {{3  $\rightarrow$  Axis}, {5  $\rightarrow$  Axis}, {7  $\rightarrow$  {1}}},
  PlotRange  $\rightarrow$  {{1, dmax}, {0.4, 1.25}},
  AxesOrigin  $\rightarrow$  {3, 0.4}, AxesLabel  $\rightarrow$  MaTeX[{"d", "\\tau"}], LabelingSize  $\rightarrow$  Full,
  Ticks  $\rightarrow$  {3 Range[5], Automatic},
  ImageSize  $\rightarrow$  Large, GridLines  $\rightarrow$  {Range[dmax - 2] + 2, None}]

```

Out[132]=



In[133]:=

Export["figtotalloppict.pdf", figtotalloppict]

Out[133]=

figtotalloppict.pdf

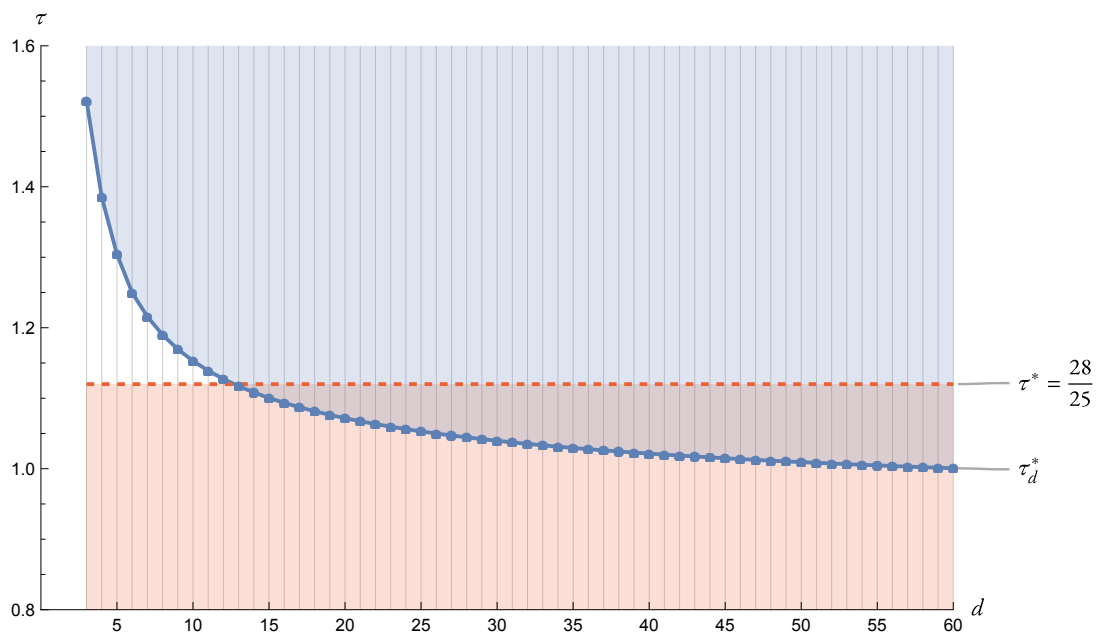
§2.3. An illustration of main results

■ Figure 1

In[159]:=

```
figtotalpictsmall = ListLinePlot[
  {Table[{d, 1.6}, {d, 3, 60}],
   Table[{d,  $\tau_{\text{star}}[d]$ }, {d, 3, 60}], Table[{d, 28/25}, {d, 3, 60}]},
  PlotMarkers → {None, {●, Offset[2]}}, None},
  PlotStyle → {None, {clr[[1]], {clr[[4]], Thick, Dashed}}},
  PlotLabels → {None,  $\tau^*$ ,  $\tau_d^*$ },
  Filling → {{2 → {1}}, {3 → Axis}},
  PlotRange → {{0, 60}, {0.8, 1.6}},
  AxesOrigin → {0, 0.8}, AxesLabel → {d,  $\tau$ }, LabelingSize → Full,
  Ticks → {5 Range[12], Automatic},
  ImageSize → Large, GridLines → {Range[58] + 2, None}]
```

Out[159]=



```
Export["figtotalpictsmall.pdf", figtotalpictsmall]
```

Out[164]=

figtotalpictsmall.pdf

■ Enclosures ensuring that Λ_d^* are upper bounds for $\tau_d^{*3} d^{3/2}$

In[167]:=

```
 $\Delta_{\text{star}} = \langle | 3 \rightarrow 19, 4 \rightarrow 22, 5 \rightarrow 25, \\ 6 \rightarrow 29, 7 \rightarrow 34, 8 \rightarrow 39, 9 \rightarrow 44, 10 \rightarrow 49, 11 \rightarrow 54, 12 \rightarrow 60 | \rangle;$ 
```

```
TableForm[Table[Flatten[{d,  $\tau_{\text{star}}[d]^3 d^{3/2}$ ,  $\Lambda_{\text{star}}[d]$ ,
  LBound[T[d,  $\Lambda_{\text{star}}[d]^{1/3} / \sqrt{d}$ ], 10-6]}], {d, 3, 12}],
TableHeadings → {None, MaTeX[{"d", " $\tau_{\text{star}}^3 d^{3/2}$ ",
  " $\Lambda_{\text{star}}$ ", "\text{lower bounds for}
  }\\mathcal{T}_d\\left(\\Lambda_{\text{star}}^{1/3}/\\sqrt{d}\\right)\\text{
  {, all positive}}"}]]]
```

Out[188]//TableForm=

d	$\tau_d^3 d^{3/2}$	Λ_d^*	lower bounds for $\mathcal{T}_d \left(\Lambda_d^{*1/3} / \sqrt{d} \right)$, all positive
3	18.3261	19	$\frac{254\,172\,122\,470\,243\,857\,187\,875}{38\,685\,626\,227\,668\,133\,590\,597\,632}$
4	21.2625	22	$\frac{260\,775\,792\,285\,720\,236\,271\,105}{38\,685\,626\,227\,668\,133\,590\,597\,632}$
5	24.8089	25	$\frac{15\,296\,388\,626\,154\,291\,952\,715}{9\,671\,406\,556\,917\,033\,397\,649\,408}$
6	28.6799	29	$\frac{45\,126\,767\,439\,274\,258\,052\,623}{19\,342\,813\,113\,834\,066\,795\,298\,816}$
7	33.2786	34	$\frac{178\,311\,456\,678\,250\,552\,577\,669}{38\,685\,626\,227\,668\,133\,590\,597\,632}$
8	38.1254	39	$\frac{96\,047\,214\,635\,786\,503\,033\,603}{19\,342\,813\,113\,834\,066\,795\,298\,816}$
9	43.1923	44	$\frac{159\,448\,793\,049\,023\,429\,729\,303}{38\,685\,626\,227\,668\,133\,590\,597\,632}$
10	48.4601	49	$\frac{48\,343\,761\,363\,350\,285\,852\,493}{19\,342\,813\,113\,834\,066\,795\,298\,816}$
11	53.9149	54	$\frac{868\,977\,098\,161\,909\,352\,477}{2\,417\,851\,639\,229\,258\,349\,412\,352}$
12	59.5457	60	$\frac{16\,952\,017\,587\,702\,127\,491\,045}{9\,671\,406\,556\,917\,033\,397\,649\,408}$

■ Figure 2

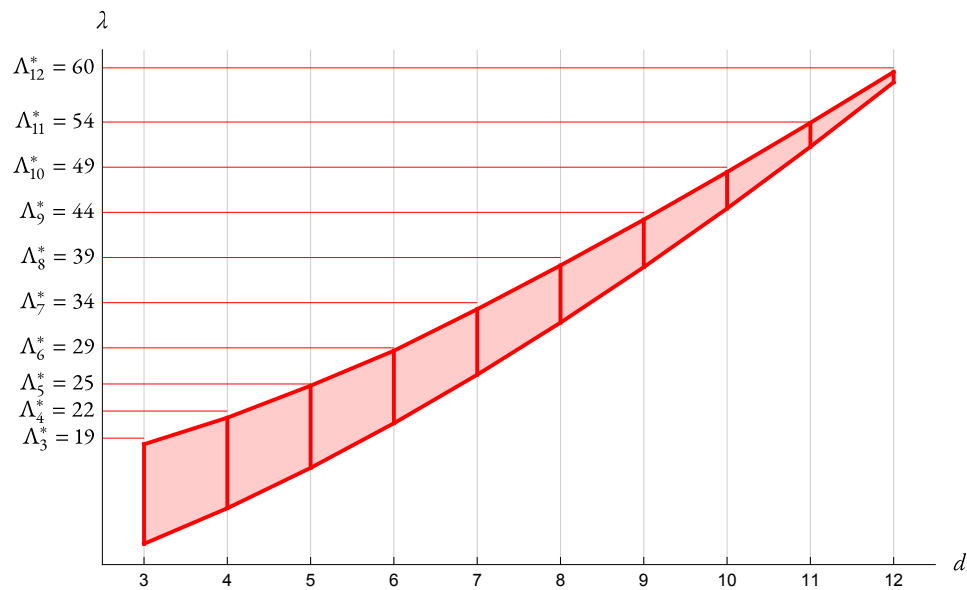
In[184]:=

```

stp = 2.5; enp = 12.5;
tp =
  Table[
    { $\Lambda^*[d]$ ,  $\text{MaTeX}["\\Lambda^*_{"} <> \text{ToString}[d] <> "]" = <> \text{ToString}[\Lambda^*[d]]$ },
    {(d - stp) / (enp - stp), 0}, Red}, {d, 3, 12}];
figlambdagap = ListLinePlot[{Table[{d,  $d^{(3/2)} \tau^*[d]^3$ }, {d, 3, 12}],
  Table[{d,  $d^{(3/2)} (28/25)^3$ }, {d, 3, 12}], PlotStyle -> {{Thick, Red}},
  PlotRange -> {{stp, enp}, {5, 62}}, AxesLabel -> {MaTeX["d"], MaTeX["\\lambda"]},
  Filling -> {1 -> {2}}, Ticks -> {Range[10] + 2, tp}, Epilog -> {Thick, Red, Table[
    Line[{d,  $d^{(3/2)} \tau^*[d]^3$ }, {d,  $d^{(3/2)} (28/25)^3$ }], {d, 3, 12}]},
  GridLines -> {Range[10] + 2, None}, Ticks -> {Automatic, tp}, ImageSize -> 500]

```

Out[185]=



In[186]:=

```
Export["figlambdagap1.pdf", figlambdagap]
```

Out[186]=

figlambdagap1.pdf

§5. Filling the gap

Extra script (should be downloaded to the same folder)

In[303]:=

```
Get["UltrasphericalDerivativeZeros.wl"] (* scripting by Claude *)
```

A single dimension, reproducing one row of Table 1

```

In[304]:=
  listP = ComputeListP[3, 19, 1/100];    (* d = 3, Lambda^*_3 = 19  *)
  Length[listP]                          (* K_3 = 50                *)
  NeumannCountingFunction[listP, 3]      (* N^Neu = 570             *)
  PolyRatio[3, ExpandListP[3, listP]] (* exact; approx 0.9061  *)
  Dataset[ExpandListP[3, listP], ItemDisplayFunction -> (Identity[#] &)]
  (* the expanded list                *)

Out[305]=
50

Out[306]=
570

Out[307]=
1 676 774 056 621 368 167
1 850 557 234 384 207 872

```

Out[308]=

n	m	k	multiplicity	lower	upper	lower(N)	upper(N)
1	0	1	1	0	0	0	0
2	1	1	3	$\frac{69\,718\,649}{33\,554\,432}$	$\frac{35\,003\,969}{16\,777\,216}$	2.07778	2.08640
5	2	1	5	$\frac{56\,070\,509}{16\,777\,216}$	$\frac{224\,639\,173}{67\,108\,864}$	3.34206	3.34738
10	0	2	1	$\frac{294\,281}{65\,536}$	$\frac{294\,665}{65\,536}$	4.49037	4.49623
11	3	1	7	$\frac{302\,896\,165}{67\,108\,864}$	$\frac{75\,829\,949}{16\,777\,216}$	4.51350	4.51982
18	4	1	9	$\frac{378\,532\,689}{67\,108\,864}$	$\frac{189\,511\,191}{33\,554\,432}$	5.64058	5.64787
27	1	2	3	$\frac{389\,129}{65\,536}$	$\frac{389\,513}{65\,536}$	5.93764	5.94350
30	5	1	11	$\frac{28\,333\,713}{4\,194\,304}$	$\frac{453\,894\,971}{67\,108\,864}$	6.75528	6.76356
41	2	2	5	$\frac{477\,713}{65\,536}$	$\frac{478\,097}{65\,536}$	7.28932	7.29518
46	0	3	1	$\frac{506\,249}{65\,536}$	$\frac{506\,633}{65\,536}$	7.72475	7.73061
47	6	1	13	$\frac{526\,265\,663}{67\,108\,864}$	$\frac{32\,930\,437}{4\,194\,304}$	7.84197	7.85123
60	3	2	7	$\frac{562\,255}{65\,536}$	$\frac{562\,639}{65\,536}$	8.57933	8.58519
67	7	1	15	$\frac{599\,437\,165}{67\,108\,864}$	$\frac{299\,890\,341}{33\,554\,432}$	8.93231	8.93743
82	1	3	3	$\frac{603\,017}{65\,536}$	$\frac{603\,401}{65\,536}$	9.20131	9.20717
85	4	2	9	$\frac{644\,829}{65\,536}$	$\frac{645\,213}{65\,536}$	9.83931	9.84517
94	8	1	17	$\frac{1\,343\,567\,715}{134\,217\,728}$	$\frac{672\,160\,207}{67\,108\,864}$	10.0104	10.0160
111	2	3	5	$\frac{695\,441}{65\,536}$	$\frac{695\,825}{65\,536}$	10.6116	10.6174
116	0	4	1	$\frac{714\,377}{65\,536}$	$\frac{714\,761}{65\,536}$	10.9005	10.9064
117	5	2	11	$\frac{725\,291}{65\,536}$	$\frac{725\,675}{65\,536}$	11.0671	11.0729
128	9	1	19	$\frac{1\,486\,918\,329}{134\,217\,728}$	$\frac{743\,868\,333}{67\,108\,864}$	11.0784	11.0845

rows 1–20 of 50

The full verification of Table 1, $d = 3, \dots, 12$

In[309]=

```

rows = GapTable[];          (* prints progress per d *)
PolyaCheckPassed[rows]      (* must be True *)
GapTableForm[rows]
TeXForm[GapTableForm[rows]] (* LaTeX source *)

```

```

d = 3,  Lambda* = 19,  K_d = 50,  N^Neu = 570,  Polya ratio = 0.906091,  < 1: True
d = 4,  Lambda* = 22,  K_d = 63,  N^Neu = 4701,  Polya ratio = 0.853325,  < 1: True
d = 5,  Lambda* = 25,  K_d = 77,  N^Neu = 36853,  Polya ratio = 0.839623,  < 1: True
d = 6,  Lambda* = 29,  K_d = 101
,  N^Neu = 379315,  Polya ratio = 0.768841,  < 1: True
d = 7,  Lambda* = 34,  K_d = 133,  N^Neu =
4307698,  Polya ratio = 0.823000,  < 1: True
d = 8,  Lambda* = 39,  K_d = 171,  N^Neu =
49018662,  Polya ratio = 0.751986,  < 1: True
d = 9,  Lambda* = 44,  K_d = 215,  N^Neu =
650176935,  Polya ratio = 0.767317,  < 1: True
d = 10,  Lambda* = 49,  K_d = 262,  N^Neu =
7562155655,  Polya ratio = 0.729007,  < 1: True
d = 11,  Lambda* = 54,  K_d = 314,  N^Neu =
105849911256,  Polya ratio = 0.748369,  < 1: True
d = 12,  Lambda* = 60,  K_d = 386,  N^Neu =
1524056239175,  Polya ratio = 0.701599,  < 1: True

```

Out[310]=

True

Out[311]//TableForm=

d	Δ^*_d	K_d	N^Neu	approx. LHS
3	19	50	570	0.9061
4	22	63	4701	0.8533
5	25	77	36853	0.8396
6	29	101	379315	0.7688
7	34	133	4307698	0.8230
8	39	171	49018662	0.7520
9	44	215	650176935	0.7673
10	49	262	7562155655	0.7290
11	54	314	105849911256	0.7484
12	60	386	1524056239175	0.7016

Out[312]//TeXForm=

```

\begin{array}{ccccc}
\text{d} & \& \text{\Lambda}^*_d & K_d & N^{\wedge}Neu & \text{approx. LHS} \\
\hline
3 & \& 19 & 50 & 570 & 0.9061 \\
\hline
4 & \& 22 & 63 & 4701 & 0.8533 \\
5 & \& 25 & 77 & 36853 & 0.8396 \\
6 & \& 29 & 101 & 379315 & 0.7688 \\
7 & \& 34 & 133 & 4307698 & 0.8230 \\
8 & \& 39 & 171 & 49018662 & 0.7520 \\
9 & \& 44 & 215 & 650176935 & 0.7673 \\
10 & \& 49 & 262 & 7562155655 & 0.7290 \\
11 & \& 54 & 314 & 105849911256 & 0.7484 \\
12 & \& 60 & 386 & 1524056239175 & 0.7016 \\
\end{array}

```

Optional float cross-check of the enclosures in a given dimension

In[313]:=

```
DiagnosticCompare[5, 25, 1/100];
```

```
77 zeros certified <= 25.; N^Neu = 36853; enclosures consistent with floats: True
```

Appendices

Appendix A

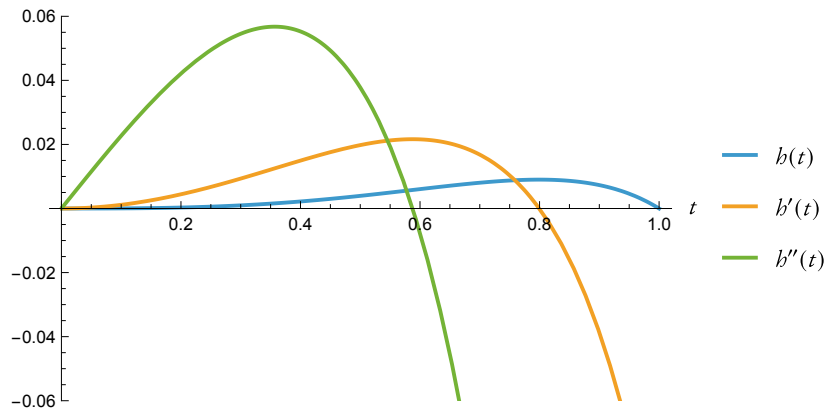
■ Figure 7

In[189]:=

```
h[t_] := Sqrt[2] t + (Pi / 2 - Sqrt[2]) t^3 - ArcCos[1 - t^2];
```

```
figphi = Plot[{h[t], h'[t], h''[t]}, {t, 0.001, 1}, PlotRange -> 0.06,
  PlotStyle -> Thick, PlotLegends -> MaTeX[{"h(t)", "h'(t)", "h''(t)"}],
  AxesLabel -> {MaTeX["t"], None}]
```

Out[190]=



In[191]:=

```
Export["figphi.pdf", figphi]
```

Out[191]=

```
figphi.pdf
```

■ Certified derivative bounds

```
h30 = Limit[h'''[t], t -> 0, Direction -> "FromAbove"] // Simplify;
```

```
h11 = h'[1] // Simplify;
```

```
h21 = h''[1] // Simplify;
```

In[208]:=

```
{ {MaTeX["h'''(0)"], h30, N[h30], "lower bound: ", LBound[h30, 0.1]},
  {MaTeX["h'(1)"], h11, N[h11], "upper bound: ", UBound[h11, 0.1]},
  {MaTeX["h''(1)"], h21, N[h21], "upper bound: ", UBound[h21, 0.1]} } //
TableForm
```

Out[208]//TableForm=

$b'''(0)$	$-\frac{13}{\sqrt{2}} + 3\pi$	0.23239	lower bound:	$\frac{312\,596\,589\,495\,787\,184\,215}{2\,361\,183\,241\,434\,822\,606\,848}$
$b'(1)$	$-2(1 + \sqrt{2}) + \frac{3\pi}{2}$	-0.116038	upper bound:	$-\frac{151\,475\,989\,880\,737\,737\,107}{9\,444\,732\,965\,739\,290\,427\,392}$
$b''(1)$	$-2 - 6\sqrt{2} + 3\pi$	-1.0605	upper bound:	$-\frac{145\,147\,172\,032\,287\,534\,964\,499}{151\,115\,727\,451\,828\,646\,838\,272}$

Appendix C.3

In[226]:=

$$\left\{ \frac{81\pi^2(-4 + \sqrt{2}\pi)}{5632}, \frac{135\pi^2(-4 + \sqrt{2}\pi)}{11264}, \frac{27\pi^2(-4 + \sqrt{2}\pi)}{11264} \right\} // \text{Reverse}$$

Out[226]=

$$\left\{ \frac{27\pi^2(-4 + \sqrt{2}\pi)}{11264}, \frac{135\pi^2(-4 + \sqrt{2}\pi)}{11264}, \frac{81\pi^2(-4 + \sqrt{2}\pi)}{5632} \right\}$$

■ Certified bounds

In[227]:=

$$\{\beta_{4,0}, \beta_{4,1}\} = \left\{ \frac{5(3\pi)^{2/3}}{28 \times 2^{1/3}}, \frac{(3\pi)^{2/3}(14\sqrt{2} + 3\pi)}{56 \times 2^{5/6}} \right\};$$

$$\{\beta_{6,0}, \beta_{6,1}, \beta_{6,2}\} = \left\{ \frac{5\pi^{4/3}}{32 \times 6^{2/3}}, \frac{\pi^{4/3}(38 + 3\sqrt{2}\pi)}{64 \times 6^{2/3}}, \frac{3^{1/3}\pi^{4/3}(6 + \sqrt{2}\pi)}{32 \times 2^{2/3}} \right\};$$

$$\{\beta_{8,1}, \beta_{8,2}, \beta_{8,3}\} = \left\{ \frac{27\pi^2(-4 + \sqrt{2}\pi)}{11264}, \frac{135\pi^2(-4 + \sqrt{2}\pi)}{11264}, \frac{81\pi^2(-4 + \sqrt{2}\pi)}{5632} \right\};$$

$$\{\beta_{4,0}, \beta_{4,1}, \beta_{6,0}, \beta_{6,1}, \beta_{6,2}, \beta_{8,1}, \beta_{8,2}, \beta_{8,3}\} // N$$

Out[230]=

$$\{0.632387, 1.30678, 0.21773, 1.11758, 1.36424, 0.0104776, 0.0523878, 0.0628653\}$$

In[248]:=

```
bdc31 = -β4,0^2 / (3 β6,0) + 1 - 4 / 9 β8,2;
bdc32 = (3 β6,0 - β6,1) + (2 / 3 β4,1 - 2 β4,0) + 1 - 4 / 9 β8,2;
bdc33 = β4,1 - β6,1;
bdc34 = β4,1 - β6,1 - 2 / 3 β8,2;
```

In[253]:=

```

TableForm[{
  {MaTeX["\\left.\\frac{\\partial\\mathcal{E}(\\sigma,\\xi)}{\\partial\\sigma}\\bigg|_{\\xi=0}"],
    MaTeX["-\\frac{\\beta_{4,0}^2}{3\\beta_{6,0}}+1-\\frac{4}{9}\\beta_{8,2}"],
    bdc31 // N, LBound[bdc31, 0.1]},
  {MaTeX["\\left.\\frac{\\partial\\mathcal{E}(\\sigma,\\xi)}{\\partial\\sigma}\\bigg|_{\\xi=\\frac{1}{3}}"],
    MaTeX["\\left(3\\beta_{6,0}-\\beta_{6,1}\\right)+\\left(\\frac{2\\beta_{4,1}}{3}-2\\beta_{4,0}\\right)+1-\\frac{4}{9}\\beta_{8,2}"],
    bdc32 // N, LBound[bdc32, 0.1]},
  {MaTeX["\\left.\\frac{\\partial\\mathcal{E}(\\sigma,\\xi)}{\\partial\\sigma^2\\partial\\xi}\\bigg|_{\\xi=0}"],
    MaTeX["\\beta_{4,1}-\\beta_{6,1}"],
    bdc33 // N, LBound[bdc33, 0.1]},
  {MaTeX["\\left.\\frac{\\partial\\mathcal{E}(\\sigma,\\xi)}{\\partial\\sigma^2\\partial\\xi}\\bigg|_{\\xi=\\frac{1}{3}}"],
    MaTeX["\\beta_{4,1}-\\beta_{6,1}-\\frac{2\\beta_{8,2}}{3}"],
    bdc34 // N, LBound[bdc34, 0.1]}
}, TableHeadings ->
{None, {"expression", "lower bound", "numerical value", "certified bound"}}]

```

Out[253]//TableForm=

expression	lower bound	numerical value	certified bound
$\left.\frac{\partial \mathcal{E}(\sigma, \xi)}{\partial \sigma}\right _{\xi=0}$	$-\frac{\beta_{4,0}^2}{3\beta_{6,0}}+1-\frac{4}{9}\beta_{8,2}$	0.364472	$\frac{9\,991\,456\,575\,844\,081\,796\,005}{37\,778\,931\,862\,957\,161\,709\,568}$
$\left.\frac{\partial \mathcal{E}(\sigma, \xi)}{\partial \sigma}\right _{\xi=\frac{1}{3}}$	$(3\beta_{6,0}-\beta_{6,1})+\left(\frac{2\beta_{4,1}}{3}-2\beta_{4,0}\right)+1-\frac{4\beta_{8,2}}{9}$	0.118743	$\frac{177\,025\,224\,130\,477\,713\,237}{9\,444\,732\,965\,739\,290\,427\,392}$
$\left.\frac{\partial \mathcal{E}(\sigma, \xi)}{\partial \sigma^2 \partial \xi}\right _{\xi=0}$	$\beta_{4,1}-\beta_{6,1}$	0.189204	$\frac{842\,505\,943\,220\,936\,596\,225}{9\,444\,732\,965\,739\,290\,427\,392}$
$\left.\frac{\partial \mathcal{E}(\sigma, \xi)}{\partial \sigma^2 \partial \xi}\right _{\xi=\frac{1}{3}}$	$\beta_{4,1}-\beta_{6,1}-\frac{2\beta_{8,2}}{3}$	0.154279	$\frac{512\,647\,013\,025\,024\,232\,459}{9\,444\,732\,965\,739\,290\,427\,392}$

Appendix C.9

- The enclosures for the RHS of (C.21), using (C.18)

In[263]:=

```

tildeQbullet[d_, τ_] :=
  Sqrt[5 d (d - 2) / (5 d^2 - 6 d - 4) - (5 (d - 2) (d + 8)) / (8 d (5 d^2 - 6 d - 4) τ^6)];
RHSC21[d_, τ_] := 193 / 77 τ^(-3)
  Exp[-2 (d - 1)^2 / (5 d (d - 2))] Exp[-(d - 1) (d + 8) / (2 (8 d^2 τ^6 - d - 8))]
  (1 - 1 / (2 τ^3 Sqrt[d] tildeQbullet[d, τ]) - 1 / (24 τ^6 tildeQbullet[d, τ]^2));

```

In[265]:=

```
TableForm[
  {{11, 4 / 5, RHSC21[11, 4 / 5] - 1 // N, LBound[RHSC21[11, 4 / 5] - 1, 0.01]},
   {16, 28 / 25, RHSC21[16, 28 / 25] - 1 // N, LBound[RHSC21[16, 28 / 25] - 1, 0.01]}},
  TableHeadings → {None,
    {d,  $\tau$ , "numerical RHS of (C.21) minus 1", "its certified lower bound"}}]
```

Out[265]//TableForm=

d	τ	numerical RHS of (C.21) minus 1	its certified lower bound
11	$\frac{4}{5}$	0.0816836	<u>1 354 065 265 920 426 872 099</u> 18 889 465 931 478 580 854 784
16	$\frac{28}{25}$	0.0113784	<u>13 018 463 748 926 440 431</u> 9 444 732 965 739 290 427 392

■ Remaining bounds

In[269]:=

```
TableForm[Table[{d, N[FdOfTau[d, IiAltf, 4 / 5] - 1],
  LBound[FdOfTau[d, IiAltf, 4 / 5] - 1, 0.01]}, {d, 4, 10}],
  TableHeadings → {None, {d, MaTeX["\\text{numerical value of }
    {}\\mathfrak{F}_d\\left(\\tau_-^{\\bullet}d^{3/2}\\right)-1"],
    "its certified lower bound"}}]
```

Out[269]//TableForm=

d	numerical value of $\mathfrak{F}_d\left(\tau_-^{\bullet}d^{3/2}\right)-1$	its certified lower bound
4	0.0728401	<u>296 754 067 571 862 444 971</u> 4 722 366 482 869 645 213 696
5	0.216595	<u>31 219 678 268 616 387 194 267</u> 151 115 727 451 828 646 838 272
6	0.31142	<u>91 098 598 266 369 414 583 455</u> 302 231 454 903 657 293 676 544
7	0.38132	<u>56 112 316 173 282 843 946 877</u> 151 115 727 451 828 646 838 272
8	0.436132	<u>64 395 206 147 717 355 855 737</u> 151 115 727 451 828 646 838 272
9	0.480841	<u>71 151 425 572 538 675 004 691</u> 151 115 727 451 828 646 838 272
10	0.518327	<u>307 264 808 301 731 429 976 683</u> 604 462 909 807 314 587 353 088

In[270]:=

```
TableForm[Table[{d, N[FdOfTau[d, IiAltF, 28 / 25] - 1],
  LBound[FdOfTau[d, IiAltF, 28 / 25] - 1, 0.01]}, {d, 4, 15}],
  TableHeadings -> {None, {d, MaTeX["\\text{numerical value of }
    \\mathfrak{F}_d\\left(\\tau_+^{\\bullet 3}d^{3/2}\\right)-1"],
    "its certified lower bound"}}]
```

Out[270]//TableForm=

d	numerical value of $\mathfrak{F}_d\left(\tau_+^{\bullet 3}d^{3/2}\right)-1$	its certified lower bound
4	0.0510292	<u>1 550 039 191 747 597 429 469</u> 37 778 931 862 957 161 709 568
5	0.0468191	<u>695 493 166 513 223 581 557</u> 18 889 465 931 478 580 854 784
6	0.0459553	<u>679 176 454 568 815 477 827</u> 18 889 465 931 478 580 854 784
7	0.0465641	<u>690 675 791 309 742 234 145</u> 18 889 465 931 478 580 854 784
8	0.0478444	<u>1 429 719 785 363 959 619 767</u> 37 778 931 862 957 161 709 568
9	0.0494237	<u>744 692 133 664 820 251 277</u> 18 889 465 931 478 580 854 784
10	0.0511118	<u>1 553 394 496 431 622 870 105</u> 37 778 931 862 957 161 709 568
11	0.0528331	<u>1 618 189 803 402 394 099 601</u> 37 778 931 862 957 161 709 568
12	0.0545201	<u>1 681 922 480 547 404 124 367</u> 37 778 931 862 957 161 709 568
13	0.056154	<u>1 743 649 743 456 745 766 967</u> 37 778 931 862 957 161 709 568
14	0.0577229	<u>1 802 922 018 977 722 577 177</u> 37 778 931 862 957 161 709 568
15	0.0592222	<u>929 781 835 317 065 459 079</u> 18 889 465 931 478 580 854 784